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$$f(x, y) = x^3 + x^2y^3 - 3y^4 \text{ in } (1, -1), \text{ graad } 2$$

$$f(1, -1) = -3$$

$$\frac{\partial f}{\partial x}(x, y) = 3x^2 + 2xy^3 \Rightarrow \frac{\partial f}{\partial x}(1, -1) = 1$$

$$\frac{\partial f}{\partial y}(x, y) = 3x^2y^2 - 12y^3 \Rightarrow \frac{\partial f}{\partial y}(1, -1) = 15$$

$$\frac{\partial^2 f}{\partial x^2}(x, y) = 6x + 3y^3 \Rightarrow \frac{\partial^2 f}{\partial x^2}(1, -1) = 4$$

$$\frac{\partial^2 f}{\partial y^2}(x, y) = 6xy^2 - 36y^2 \Rightarrow \frac{\partial^2 f}{\partial y^2}(1, -1) = -42$$

$$\frac{\partial^2 f}{\partial x \partial y}(x, y) = \frac{\partial^2 f}{\partial y \partial x}(x, y) = 6xy^2 \Rightarrow \frac{\partial^2 f}{\partial x \partial y}(1, -1) = \frac{\partial^2 f}{\partial y \partial x}(1, -1) = 6$$

$$\begin{aligned} T_2(x, y) &= f(a, b) + \frac{\partial f}{\partial y}(a, b)(y - b) + \frac{\partial f}{\partial x}(a, b)(x - a) \\ &+ \frac{1}{2} \left[\frac{\partial^2 f}{\partial x^2}(a, b)(x - a)^2 + \frac{\partial^2 f}{\partial x \partial y}(a, b)(x - a)(y - b) + \frac{\partial^2 f}{\partial y^2}(a, b)(y - b)^2 \right] \\ &= -3 + (x - 1) + 15(y + 1) + \frac{1}{2} [4(x - 1)^2 + 2(6)(x - 1)(y + 1) - 42(y + 1)^2] \\ &= -3 + (x - 1) + 15(y + 1) + 2(x - 1)^2 + 6(x - 1)(y + 1) - 21(y + 1)^2 \\ &= 2x^2 - 21y^2 + 6xy + 3x - 33y - 14 \end{aligned}$$

References